# Retail Demand Forecasting using Walmart Store Sales Data

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## 1. Overall Process Flow

Raw Data
(train.csv, features.csv, stores.csv)
↓
Data Loading
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Data Preprocessing
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Exploratory Data Analysis
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Feature Engineering
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Time Series Data Splitting
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Model Training
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*Model Explainability*
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*Business Recommendations*

## 2. Introduction

## Objective

The objective of this project is to develop a machine learning-based demand forecasting system that predicts weekly sales for Walmart stores. Accurate demand forecasting helps retailers optimize inventory, improve supply chain planning, reduce stockouts, minimize overstock situations, and support data-driven business decisions.

The forecasting model uses historical sales data along with external factors such as holidays, markdown promotions, fuel prices, temperature, Consumer Price Index (CPI), unemployment rate, and store information to estimate future weekly sales.

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## 3. Data Ingestion

## Description

The project begins by loading the datasets provided by Walmart.

The datasets include:

- **train.csv** – Historical weekly sales data
- **features.csv** – Economic indicators and promotional information
- **stores.csv** – Store characteristics such as store type and size

The datasets are loaded using the Pandas library.

Data Merging

The datasets are merged using:

- Store
- Date
- Is Holiday

Store information is merged using the Store column.

This creates a unified dataset containing sales information, external economic variables, promotional data, and store attributes.

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## 4. Data Cleaning & Transformation

The merged dataset is cleaned before model development.

## Missing Values

The MarkDown1 MarkDown5 columns contain missing values because promotional markdown information is unavailable for certain weeks. These missing values are replaced with zero, assuming no promotional markdown was applied.

Missing values in CPI and Unemployment are checked after merging. Since the merged training dataset does not contain missing values for these columns, no further imputation is required.

## Date Conversion

The Date column is converted into datetime format to enable time-based feature extraction.

## Duplicate Removal

Duplicate records are checked and removed if found.

## Data Validation

Data types are verified to ensure numerical variables are correctly interpreted by machine learning models.

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# 5. Exploratory Data Analysis

EDA is performed to understand the dataset before model development.

The following analyses are conducted:

- Distribution of Weekly Sales
- Weekly Sales Trend
- Sales by Store
- Sales by Department
- Holiday vs Non-Holiday Sales
- Correlation Analysis
- Temperature vs Sales
- Fuel Price vs Sales
- CPI vs Sales
- Unemployment vs Sales

### Key Findings

- Weekly sales exhibit strong seasonal patterns.
- Sales increase significantly during holiday periods.
- Sales vary considerably across stores and departments.
- Promotional markdowns influence sales positively.
- Historical sales demonstrate recurring demand patterns.

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## 6. Feature Engineering

Feature engineering improves the predictive capability of machine learning models.

The following features are created:

### Date Features

- Year
- Month
- Week
- Quarter
- Day

These features capture seasonal patterns.

### Promotional Features

A Total_MarkDown feature is created by summing all markdown variables.

### Lag Features

Historical sales are incorporated using:

- Lag 1 Week
- Lag 2 Weeks
- Lag 4 Weeks

Lag features enable the model to learn temporal dependencies.

### Rolling Statistics

Rolling Mean

Rolling Standard Deviation

These features summarize recent sales trends.

### Store-Level Features

Average sales for each store are calculated.

### Department-Level Features

Average sales for each department are calculated.

### Interaction Features

Additional interaction features include:

- Temperature × Fuel Price
- Holiday × Total Markdown

These capture relationships between multiple variables.

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## 7. Model Development

Three machine learning models are implemented.

### Linear Regression

Linear Regression serves as the baseline forecasting model due to its simplicity and interpretability.

### Random Forest Regressor

Random Forest captures nonlinear relationships using an ensemble of decision trees.

### XGBoost Regressor

XGBoost is selected as the advanced forecasting model because of its ability to handle nonlinear patterns, feature interactions, and large structured datasets efficiently.

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## 8. Hyperparameter Tuning

Hyperparameter tuning is performed on the XGBoost model using RandomizedSearchCV.

The tuning process optimizes parameters such as:

- Number of estimators
- Learning rate
- Maximum tree depth
- Subsample ratio
- Column sampling ratio

TimeSeriesSplit cross-validation is used to preserve chronological ordering and prevent data leakage.

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## 9. Model Evaluation

The models are evaluated using multiple regression metrics.

### Mean Absolute Error (MAE)

Measures the average prediction error.

### Root Mean Squared Error (RMSE)

Penalizes larger prediction errors more heavily.

### Mean Absolute Percentage Error (MAPE)

Measures percentage error between predicted and actual sales.

### Weighted Mean Absolute Error (WMAE)

Used because holiday weeks are more important for retail forecasting.

Holiday observations receive five times higher weight.

### $R^2$ Score

Measures how well the model explains the variance in weekly sales.

Residual analysis is also performed to verify that prediction errors are randomly distributed.

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## 10. Model Explainability

Model explainability improves trust and transparency.

### SHAP

SHAP identifies the global importance of each feature and explains how individual features influence predictions.

Visualizations include:

- SHAP Summary Plot
- Feature Importance Plot
- Waterfall Plot

## LIME

LIME explains individual predictions by identifying the features contributing most to specific forecasts.

Both techniques help business stakeholders understand the model's decision-making process.

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## 11. Business Recommendations

Based on the forecasting model, the following recommendations are proposed:

- Increase inventory before major holiday periods.
- Utilize promotional markdowns strategically to maximize sales.
- Monitor historical sales trends to improve demand forecasting.
- Tailor inventory planning based on store size and department demand.
- Use economic indicators to support long-term demand planning.

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## 12. Conclusion

This project successfully developed a comprehensive retail demand forecasting system using machine learning techniques.

After comparing multiple forecasting models, XGBoost demonstrated the best predictive performance due to its ability to capture complex nonlinear relationships and temporal demand patterns.

The integration of SHAP and LIME provides transparent model explanations, making the forecasting system suitable for business decision-making. The proposed solution can help retailers optimize inventory, improve promotional planning, and enhance overall supply chain efficiency.